Applying Natural Language Processing to Protest Event Analysis

A Case Study of UK Media Before and After COVID-19

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Abstract

This project applies Natural Language Processing (NLP) to automate key parts of Protest Event Analysis (PEA) using UK news coverage before, during, and after the COVID-19 period. Using MediaCloud, protest related candidate articles between 2020 and 2024 from three UK outlets were collected. The pipeline then scraped full article text and stored them.

To focus on articles whose main topic is a concrete protest event, this study uses a transformer based zero-shot classifier to assign each article a `PROTEST` or `NOT_PROTEST` label. For the protest subset, the project applies sentiment analysis, exploratory topic modelling to summarise recurring themes, and temporal analysis to examine how protest coverage changes over time and across outlets.

Overall, the results shows that an end to end automated workflow can build and manage a protest corpus at scale, while still requiring careful validation and transparent reporting of errors and data loss. The analysis suggests that protest reporting is concentrated in a small number of high attention periods and that patterns of coverage and themes vary across outlets and across phases of the pandemic timeline.

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1 Introduction

Protest Event Analysis (PEA) is a research method used in social and political science to find, label, and study protest activity using media reports. It means turning unstructured texts like newspaper articles into structured details, like date, place, demands, and organisations, so researchers can study patterns in collective action [Hutter, 2014].

1.1 Research context: protest, COVID-19 and UK media

Protest is one of the clearest ways that ordinary people take part in politics. Demonstrations, strikes, occupations, and other forms of group action show what people think is unfair, what issues they want politicians to address, and how they judge governments and other powerful actors. Because protesters often have limited power, protests are an important tool for understanding whose voices are heard in politics.

In this wider area, Protest Event Analysis (PEA) is a key approach. In practice, PEA turns newspaper reporting into structured event records like date, place, actors, form of action, and demands. This created the possibility of analyse protest patterns over time and compare coverage across outlets. This event based approach is used both in classic PEA research and in large datasets such as ACLED, which also record the date, place, actors, and type of protest or political violence [ACLED, 2015]. By turning unstructured text into event records that can be analysed, PEA makes it possible to track protest patterns over time, compare issues and groups across countries, and connect protest waves to changes in politics or socioeconomic conditions.

The COVID-19 pandemic quickly challenged this kind of "street politics". Lockdowns, curfews, and bans on gathering strongly limited traditional demonstrations, especially early on. Studies in Spain, for example, show that even though overall protest participation dropped and protests became smaller, shorter,

and more limited in space, the number of protest events in 2020 did not fall as much as many would expect [Romanos et al., 2022]. Globally, data based research suggests that COVID-related disruption, together with existing problems like financial insecurity and low trust in institutions, influenced which places saw protests and at what times [van der Zwet et al., 2022]. So rather than stopping protest, COVID-19 changed who could protest, how they protested, and which issues became central. Importantly, the pandemic also created new protest issues and new alliances. In many countries, movements formed that directly opposed public health measures: they protested against lockdowns, mask rules or vaccination campaigns and described these policies as threats to personal freedom and control over the body. At the same time, other activists linked COVID-19 to older problems like insecure work, racial injustice, housing issues, or underfunded public services, arguing that the crisis revealed and worsened existing inequalities[Rohlinger and Meyer, 2022]. Comparative research on anti-containment protests in Germany and Italy, for example, shows that far right groups and alternative communities played a major role, and that protest messages and styles became more confrontational and more linked to conspiracy ideas over time, even though mobilisation developed differently in each country [della Porta and Hunger, 2024]. In the UK, anti lockdown and anti vaccine marches became a visible part of protest politics, alongside broader protests linked to the social and economic effects of the pandemic and to rights based issues.

Because most people do not see protests directly, news media are the main way protests become visible to the public. For decades, PEA research has relied on newspapers because they offer a relatively accessible record of when and where protests happen, who takes part, and what people demand. Media reports therefore work as a practical source for rebuilding the "when, where, who, and what" of protest over long periods. At the same time, newspapers do not completly reflect reality. Editorial choices about what to cover and how to describe it shape which protests become visible and how they are understood.

This project focuses on the UK media context. Substantively, the UK press is large and politically diverse, from left leaning broadsheets to right wing tabloids. This makes it a useful case for studying how protest is selected and framed across the political spectrum, especially before and after the major disruption of COVID-19. However, UK newspapers do not only report on protests inside the UK. They also cover protests around the world, choosing which international events to highlight and how to explain them to the UK public. For this reason, this dissertation studies how UK based media outlets represent protest events globally, rather than focusing only on protests within the UK. This design follows the main idea of Protest Event Analysis: coding protest information in a systematic way using media sources. It also lets us study how UK newspapers report and frame

protests across different topics, places, and time periods during COVID-19

1.2 Research problem and motivation

Doing protest event analysis manually from news articles has several practical and methodological problems. First, it can be difficult to access enough news data: many newspapers are behind paywalls, limit web scraping, or block automated collection for technical or legal reasons. In practice, this forces researchers either to use only a small set of accessible outlets, or to collect articles by hand, which takes a huge amount of time. Second, newspapers often report details in inconsistent ways. There is no single, clear "ground truth" in news reporting, so researchers often need to read multiple articles about the same event and still make judgement calls about what to trust. Together, limited access, inconsistent reporting, and detailed manual coding make news based PEA slow and hard to scale to today's volume of digital news.

These limits point to a clear practical and academic gap. Because access is restricted and reporting is inconsistent, manual PEA often produces small, incomplete datasets that are difficult to compare. This makes it harder to do strong long term analyses and harder to compare coverage across newspapers with different political positions. In short, there is a mismatch between the huge amount of digital news available and the smaller, biased protest datasets produced by current manual methods. This dissertation addresses that gap by building an end to end automated pipeline that collects articles from multiple UK outlets and classifies which ones report protest events.

Using NLP and, more recently, large language models to automate parts of PEA is promising because it targets the main bottlenecks in manual coding. Text classification and information extraction models can filter large news archives, flag likely protest reports, and suggest key details about events. That reduces the amount of documents that human coders must read. For example, an NLP pipeline could reduce a corpus of millions of news articles to a smaller set of protest relevant texts, while leaving final event coding to humans. This would make it more realistic to build protest datasets that cover many years and many sources in a systematic and reproducible way. However, automated approaches bring their own risks. Fully automated protest and conflict datasets can include out of context coding, misclassification, many duplicated events and many pipelines are not transparent enough for researchers to properly assess their biases. NLP and LLMs are trained on biased text and can reproduce the biases of the media and society. If treated as neutral tools, they may reinforce these problems rather

than reduce them. For these reasons, recent work supports hybrid approaches: NLP tools handle large scale search and filtering, while human coders remain essential for dealing with ambiguity, validating events, and checking data quality [Lorenzini and Kriesi, 2021].

Although classic Protest Event Analysis often codes event attributes such as location, actors and demands, this dissertation does not implement full event level information extraction. In this corpus, many articles do not report these attributes in a consistent or complete way; they often focus instead on developments around protests (for example, political responses, policing decisions, disruption, or clashes) rather than providing a standardised event summary. Building and validating a reliable extraction system would therefore require substantially more manual annotation and case by case interpretation than is feasible within this project. Instead, the dissertation focuses on automating protest detection at the article level and then analysing protest coverage using sentiment analysis and topic modelling to compare tone and themes across outlets and over time, including coverage of protest events reported in the UK press.

Taken together, these challenges motivate the contribution of this dissertation: a reproducible NLP pipeline that can scale the early stages of Protest Event Analysis while remaining transparent and open to critical evaluation. The next chapter sets out the aims, research questions, and objectives that guide the design of this pipeline and its application to ideologically diverse UK newspapers across the COVID-19 period and its aftermath.

1.3 Structure of the report

This report is organised into a sequence of chapters that follow the main steps of the project. After this introduction, Chapter 2 sets out the research aims, objectives, and research questions. Chapter 3 reviews the key literature on protest event analysis, media reporting, and NLP approaches. Chapter 4 describes the data sources and how the dataset was built. Chapter 5 explains the methods and the end to end NLP pipeline used to identify protest articles and analyse them. Chapter 6 summarises the technical implementation and how the code and outputs are organised. Chapter 7 reports the evaluation and validation of the classifier and key design choices. Chapter 8 presents and discusses the main results, comparing patterns across time periods and newspapers. Finally, Chapter 9 concludes by summarising the main findings, noting limitations, and suggesting future work.

2 Aims and Objectives

2.1 Aims and objectives

The aim of this dissertation is to build and critically evaluate a reproducible NLP pipeline that automates key steps of PEA using street protest reporting in UK newspapers, and then use this pipeline to compare how protests are reported across ideologically different outlets from 2020 to 2024. In practical terms, the project moves from raw online news content to a structured, analysable dataset by combining; large scale article collection, protest relevance classification and later text analysing sentiment and topics. The pipeline is applied consistently to three UK titles that span the political spectrum so that differences in protest coverage can be studied across outlets and across time, with particular attention to the COVID-19 period and its aftermath.

The project is guided by three research questions:

- How effectively can an automated NLP pipeline distinguish genuine protest reports from other news articles in a large corpus?
- How does the sentiment and framing of protest coverage vary across ideologically diverse newspapers and over time?
- To what extent do topic patternsd in protest reporting shift across the pandemic timeline?

To answer these questions, the work is organised around a small set of connected objectives that together form an end to end workflow. First, the study defines the problem clearly by reviewing research on PEA, COVID-19 protest dynamics, media framing, and automated event data methods. Then, it turns this review into clear design requirements for a pipeline that is scalable and transparent.

Second, it constructs the empirical base: a 2020–2024 corpus of news stories gathered through MediaCloud and outlet specific scraping, stored in a queryable database with a documented schema.

Third, it develops and evaluates a binary protest classifier to identify which articles describe real world protest events. This includes creating a manually labelled sample for evaluation, selecting a suitable model, tuning the decision threshold, and reporting performance using standard metrics like accuracy, precision or recall, before writing predictions back into the database for the full corpus.

Fourth, the dissertation analyses how protests are portrayed in the predicted protest subset. Sentiment analysis is used to compare the tone of protest coverage by outlet and period, while topic modelling is used to identify recurring issue areas and track how these change over time and differ across newspapers. Where possible, the project checks the reliability of these models through basic diagnostics.

Finally, the pipeline outputs are used for comparative analysis and visualisation. The results are presented as clear time series and distributional comparisons of protest volume, sentiment, and topic prevalence across newspapers and across periods. All steps are implemented as reusable code with enough documentation to support replication, extension, and critical evaluation by other researchers.

3 Background and Literature Review

3.1 Problem description

This dissertation tackles a practical problem in Protest Event Analysis: how to study protest reporting at scale when the main sources are large amounts of newspaper text. To examine how UK press coverage of protest changed across the COVID-19 period, the project needs a reproducible NLP workflow that can:

- Identify which articles describe real world protest events
- Make comparisons that are consistent across different newspapers and time periods

This chapter reviews the key research that motivates these choices and highlights the main challenges: bias in what news outlets cover, missing or unclear event details, and the limits of automated text analysis.

3.2 Protest Event Analysis: concepts and methods

At its most basic, a PEA study proceeds in three steps. First, the researcher defines the population of texts to be analysed, such as one or more newspapers over a given period. Second, coders read each text and identify mentions of protest events according to an explicit definition of what counts as a protest. Third, each event is coded on a series of dimensions using a standardised codebook, producing a rectangular dataset where each row is an event and each column is an attribute. This approach is attractive because it allows researchers to link the theoretical language of social movement research to concrete variables that can be quantified and compared.

Within this framework, several concepts are central: events, claims, repertoires, frames and actors. They provide the bridge between substantive theories

of contention and the variables that PEA typically records.

The event is the primary unit of analysis in PEA. An event is usually defined as a discrete episode of public, collective action in which a group of actors expresses a political or social claim in a contentious way, for instance through a demonstration, strike, blockade, occupation or rally. For each event, coders typically record attributes such as: date, location, size, form of action, the presence of violence or property damage, the occurrence of arrests or police intervention, and any immediate outcomes.

Because events are taken from media reports, they reflect both protest dynamics on the ground and editorial choices about what is newsworthy. Nonetheless, event level data make it possible to produce time series of protest activity, compare issue fields, and study the diffusion or decline of particular forms of contention.

Beyond counting events, PEA is often concerned with the claims articulated by protesters: the problems they define, the demands they make and the targets they address. Building on this idea, Koopmans and Statham (1999) developed political claims analysis, which integrates protest events with non protest statements in the public sphere. In this approach, each act of contention or public action is coded by its issue area, such as welfare, immigration, climate, or civil liberties. It is also coded by what is described as the problem, what solution is proposed, and sometimes who is blamed.

In practice, many PEA datasets use keywords to capture claims, while more recent work experiments with automated text analysis to classify issues and detect frames. In the context of this thesis, topic modelling and manual coding of issue categories can be seen as an approximate implementation of this claims dimension: they help to identify which grievances and demands dominate protest coverage in different periods and outlets.

The concept of repertoires of contention, associated above all with Charles Tilly, refers to the set of protest forms that are available and meaningful to actors in a given time and place [Tilly, 1993]. Rather than inventing tactics from scratch, movements tend to draw on a limited menu of familiar "performances" such as marches, strikes, sitins, boycotts, petitions or, more recently, online campaigns. These repertoires evolve historically as political opportunities, technologies and organisational forms change.

PEA turns into clear repertoires by coding the form of action used in each event for example, demonstration, strike, blockade, occupation, riot, and sometimes by distinguishing degrees of disruption or violence. This allows researchers to ask not only how much people protest, but also how they protest, and whether certain

periods are associated with shifts towards more dispersed, digital or innovative tactics

Protests are not only about resources and opportunities, but also about meaning. Framing is how people explain what the problem is, who caused it, and what to do about it. Benford and Snow, for instance, distinguish problem definition, solutions and motivation [Benford and Snow, 2000].

In PEA, frames can be approached from two sides. Movement frames concern how protesters themselves describe their cause, often captured through slogans and quotes in news coverage. Media frames concern how journalists and outlets present the protest. While traditional PEA typically codes only basic information about issues, some studies extend the method to capture frames more explicitly, either through additional manual categories or through automated sentiment analysis. In this thesis, sentiment analysis of protest related articles and comparisons of tone across newspapers are used as one way of approximating media frames.

The literature highlights several strengths of traditional, manually coded PEA. First, it is a systematic and transparent method: codebooks define clearly which events are included and how variables are recorded, which facilitates replication and comparison across studies [Hutter, 2014]. Second, because newspapers publish daily and maintain archives, PEA can produce large datasets that cover multiple decades and allow the analysis of long term trends in contention. Third, news reports often contain rich contextual information, quotes, descriptions of the setting, references to previous episodes, which enables researchers to go beyond simple event counts and examine claims, repertoires and interactions in detail.

At the same time, PEA has well known limitations. The most frequently discussed is that manual coding is extremely labour intensive and time consuming. Constructing a sizable dataset usually requires teams of coders reading and coding thousands of articles, which is costly and difficult to scale across many countries, outlets or long periods. This has led some authors to call for methodological innovations to reduce the reliance on human coding [O'Brien, 2018].

A second limitation arises from the reliance on media sources. Newspapers introduce both selection and description biases. They do not report every protest, but tend to focus on events that are large, dramatic, violent or geographically close to their readers. Also they may under report or ignore actions by marginalised groups or in peripheral locations. Even when protests are covered, information on key attributes such as claims, actors or size may be incomplete or skewed, and reporting often relies heavily on official sources such as the police.

Third, because coding is carried out by humans, PEA is subject to coder

fatigue, error and subjectivity. Most projects attempt to mitigate this through training and inter coder reliability tests, but some degree of inconsistency is difficult to avoid. Finally, PEA based solely on mainstream newspapers may struggle to capture small scale, everyday or online forms of contention that fall below the threshold of traditional news coverage.

These strengths and weaknesses provide an important context for the present project. On the one hand, the thesis follows the PEA tradition in treating protest events, claims, repertoires, frames and actors as key dimensions of analysis, and in using newspaper coverage as a primary source for reconstructing protest dynamics internationally. On the other hand, it responds to the limitations of manual PEA by exploring how natural language processing and machine learning can automate parts of the coding process. It reduce the volume of human labour required, and make it easier to scale protest event analysis across larger datasets and longer time periods, while remaining attentive to the persistent biases of media based data.

3.3 Protests before, during and after COVID-19

Research on protest during COVID-19 finds that protest activity dropped at the beginning of the pandemic, when lockdowns and uncertainty made mobilising difficult, but the decline did not last long. Using global event data from May 2020 to July 2022, Hellmeier (2023) finds that protest activity quickly rebounded and that, on average, around 15% of all recorded protests in this period were directly related to the pandemic, with a peak of more than 50% in April 2020. In person demonstrations remained common despite public health restrictions, and pandemic related contention clustered around seven main issue areas: masks and vaccination, economic hardship, business restrictions, health care, education, government mismanagement, and crime.[Hellmeier, 2023] Descriptive accounts of anti lockdown and anti mandate protests likewise underline how "classic" street repertoires; marches, rallies, occupations, and, in some cases, riots were adapted rather than abandoned, including socially distanced protests and motorcades.

Within the 2020–24 window of this study, several overlapping protest waves stand out. First, the Black Lives Matter mobilisations after the murder of George Floyd in May 2020 triggered one of the largest recent protest waves in the US and sparked solidarity protests in the UK. Second, from March 2020 into late 2022, many countries saw repeated protests against lockdowns, masks, and vaccines, as well as "freedom convoy" style actions. Around the same time, political polarisation also shaped protest agendas, especially after the January 2021 US

Capitol assault, which fuelled wider debates about democracy, extremism, and public order. From late 2021 onwards, rising prices and economic strain drove a new wave of cost of living protests, while Russia's invasion of Ukraine in February 2022 prompted anti war and solidarity demonstrations and intensified energy related grievances. Finally, the war in Gaza from late 2023 generated large and sustained protest activity in the UK, particularly around calls for a ceasefire and humanitarian access.

3.4 Media bias and ideological differences in protest reporting

It is well established that newspapers differ in their political orientation, and this shapes both which protests they report and how they report them. In commercial media markets, editors prioritise events that are likely to attract attention, so larger, more dramatic or conflictual protests are more newsworthy than small, routine actions. This produces selection bias: some protests never appear in the news at all, while others receive disproportionate coverage. Once a protest is selected, additional bias can enter through tone and framing, for example, emphasising violence, disruption and police viewpoints versus highlighting grievances, rights and democratic participation.

Public opinion data from YouGov show that British readers perceive a clear ideology in the national press: The Guardian is seen as the most left wing mainstream newspaper, while The Daily Mail is viewed as the most right wing [Smith, 2017]. This project focuses on three UK titles that covers this spectrum and are accessible in MediaCloud: The Guardian (centre left), The Daily Mail (right wing), and The Evening Standard (right leaning). B.1

Comparative research from other contexts gives an idea of how strongly newspapers ideology can shape protest coverage. Susánszky et al. (2022) analyse leading Hungarian online news portals and find little difference between progovernment and governmentcritical media in whether protests are reported, but sharp differences in how they are described. Progovernment outlets systematically deploy protest paradigm devices, a derogatory tone and populist "enemy" narratives that portray protesters as illegitimate or manipulated, while critical outlets are more likely to foreground grievances and democratic rights [Susánszky, 2022]. This suggests that political bias in protest coverage is often not about ignoring protests, but about how they are described and presented.

4 Data

This chapter describes the data used in the project. The dissertation first introduces the case study and selected newspapers, then explains how candidate articles were collected from MediaCloud and scraped into full text. Finally, there is a description of the structure of the MongoDB collections that store the corpus and the manually labelled validation sample.

4.1 Case study overview: UK newspapers and protest reporting

The empirical focus of this study is protest reporting in three major UK news outlets between 1 January 2020 and 31 December 2024. The original plan was to include titles such as The Times and The Daily Telegraph in order to cover a wider range of broadsheet outlets. In practice, these had to be excluded because their web content could not be scraped reliably or without breaching terms of use, whereas the three chosen outlets could be accessed with the tools available for this study.

The choice of outlets allows a clear political comparison. The Guardian is generally left of centre, The Daily Mail is right wing, and The Evening Standard is slightly right leaning with a strong focus on London. What is missing is a genuinely centrist newspaper and more radical or explicitly far left or far right media. As a result, the study compares how mainstream left and right newspapers cover protest, rather than mapping the full ideological spectrum of the UK media system.

4.2 Data sources

URLs were retrieved from MediaCloud and scraped into full text; the full pipeline is summarised in Section 5.1

4.2.1 Time frame and inclusion criteria

The current dataset covers the period from **2020-01-01** to **2024-12-31**. This window was chosen because it spans three analytically important phases:

- A short pre pandemic period in early 2020, before COVID-19 restrictions were fully in place.
- The pandemic period itself, with lockdowns, social distancing rules and COVID-related protest.
- A post pandemic phase in which public health restrictions were lifted but other issues generated new waves of contention.

Articles were considered "in scope" according to the following criteria:

- **Language and geography:** The MediaCloud collections *United Kingdom – National* and *United Kingdom – State & Local* were used, and only records with English language metadata were retained. These MediaCloud collections refer to UK based media sources.
- **Protest-related keywords in MediaCloud:** MediaCloud was queried for articles whose indexed fields contain at least one of the following terms: `''protest''`, `''protests''`, `''riots''`, or `''protesters''`.
- **Selected outlets:** From the resulting CSV export, only records where `media_url` matched one of the three target outlets were kept: `theguardian.com`, `dailymail.co.uk` and `standard.co.uk`.
- **News-focused sections and titles:** Obvious non news sections such as sport, lifestyle or culture were excluded based on URL prefixes and simple title rules.

This means that the dataset does not include every article from these newspapers, but only those that contain protest related keywords in the MediaCloud index and pass the additional cleaning steps.

4.2.2 MediaCloud metadata

MediaCloud was used to download basic metadata for candidate articles in CSV format, in the `URL.csv` file. The fields used in this project are:

- `id`: a unique MediaCloud identifier for the article.
- `indexed_date`: the timestamp when MediaCloud indexed the article.
- `language`: the language of the article.
- `media_name`: the name of the media outlet.
- `media_url`: the domain of the outlet.
- `publish_date`: the publication date.
- `title`: the article headline as stored by MediaCloud.
- `url`: the URL of the article on the outlet's website.

The Python script `FirstQuery.py` performs the initial cleaning and filtering. The main steps are:

1. **Load and rename columns.** The CSV is read using `pandas`, rows beyond the header artefacts are dropped, and the columns are renamed to the eight fields above.
2. **Filter to the three outlets.** Records are kept only if their `media_url` corresponds to one of the three newspapers. At this point, only the columns `id`, `media_url`, `publish_date`, `title` and `url` are retained.
3. **Exclude non news sections by URL prefix.** A list of URL prefixes is defined for each outlet that corresponds to sections such as sport, lifestyle, culture, travel, comment, magazines, galleries and similar. Any article whose URL starts with one of these prefixes is removed. This helps restrict the dataset to news and related content, rather than, for example, travel guides or celebrity features.
4. **Exclude generic headline and briefing pages.** A small helper function `is_excluded_title` flags titles that clearly correspond to round ups rather than single articles, such as those starting with "News headlines" or "Morning headlines", titles containing patterns such as "*[weekday] briefing: ...*", or generic items like "photos of the day". These are also removed.
5. **Save cleaned URLs.** The resulting set of candidate articles for scraping is written to a file `URLs_clean.csv`.

4.2.3 Web scraping of article text

The second stage of data collection scraped full article text for the URLs in `URLs_clean.csv`. The final pipeline uses `newspaper3k` (`from newspaper import Article, Config`) to download pages and extract the main body text.

Overall, 12,506 candidate URLs were stored in MongoDB and 12,496 were successfully scraped. Each URL is assigned a `status` value so unsuccessful items can be identified. Items with missing or very short text were handled in later stages.

4.3 Data schema and storage (MongoDB)

After scraping, classification and basic filtering, the data are stored in a MongoDB database with two collections:

- `Texts`: the main corpus of 12,506 articles, including both successfully and unsuccessfully scraped items.
- `sample_texts`: a manually labelled subset of 85 articles used for validation.

Using MongoDB allows the project to store raw text and derived fields together, while keeping the schema flexible as new variables, are added. To have an independent check on the automatic protest classification, a smaller validation set with 85 random articles was created.

The `Texts` collection contains one document per URL. Its structure is summarised in Table C.1.

The article URL is used as the `_id` field so that each document is uniquely identified without introducing a separate numeric key. This simplifies linking data across different processing steps and avoids duplicates. The combination of `status` and `hf_status` allows the pipeline to distinguish problems in scraping from problems in classification.

The texts that are defined as "PROTEST" have the `sentiment` field, with the schema in this Table C.2.

The `sample_texts` collection stores these 85 articles, with the schema shown in Table C.3.

The manual labels in this collection are imbalanced: 62 of the 85 articles are labelled as `0` (not protest) and 23 as `1` (protest). Even though it is imbalanced, it provides a useful small scale benchmark for assessing how well the final classification pipeline distinguishes protest from non protest content.

Together, the `Texts` and `sample_texts` collections form the core data infrastructure for the project: a relatively large corpus of protest related news articles with machine generated labels, and a smaller manually coded subset for validation and error analysis.

4.3.1 Data limitations

The dataset is useful for analysing protest coverage, but it has several clear limitations. First, it only includes three outlets:The Guardian, The Daily Mail and The Evening Standard. This gives a left–right contrast within mainstream news, but excludes other major and regional titles, as well as more centrist or more partisan media. The findings should therefore be read as patterns in these three newspapers, not the whole UK media system.

Second, the data depend on MediaCloud's coverage and on keyword based queries. MediaCloud indexes only content available on the open web and can miss articles behind strict paywalls, in apps or in formats it cannot parse. The use of protest related keywords plus a wider term list in the scraped text means that articles describing protests without this vocabulary are likely to be missed, while some pieces that mention protest only briefly are included.

Third, web scraping and text cleaning are imperfect. Although only a small number of URLs fail completely, some scraped texts may still contain some errors or short fragments, because the project relies on `newspaper3k`'s extraction and a simple length filter. This introduces some noise into the input for classification and any later NLP steps.

Finally, the manually labelled validation set is small and imbalanced. It is useful for basic checks on the automatic labels, but it does not provide a large, carefully sampled test set. Combined with the temporal restriction to 2020–2024, these factors mean that later results should be interpreted as indicative patterns within this specific corpus, rather than definitive statements about all protest reporting.

5 Methodology and Methods

This chapter sets out the overall research design and the NLP pipeline used to analyse UK newspaper coverage of protest between 1 January 2020 and 31 December 2024. The project studies how newspapers write about protest, rather than trying to reconstruct a definitive list of protest events.

The chapter is organised as follows. Section 5.1 presents the end to end workflow from MediaCloud queries to later analyses, and distinguishes between engineering and experimental components. Section 5.2 describes the binary protest vs. non protest classification task and the use of a zero-shot classifier. Sections 5.3–5.5 outline the design for sentiment analysis, topic analysis, and temporal analysis. Finally, Section 5.6 summarises the evaluation strategy across the different pipeline components.

5.1 Overview of the NLP pipeline

End to end workflow

The project implements an end to end workflow that turns a large set of candidate news URLs into a structured protest corpus for analysis. In brief, the pipeline has four stages:

1. **MediaCloud retrieval and filtering.** MediaCloud is used to collect English language candidate stories from UK collections using protest related keywords. The export is then filtered to the three target outlets and obvious non news sections are removed. Data collection and inclusion criteria are described in Chapter 4.

2. **Scraping and storage.** Candidate URLs are scraped to extract headline and article text, and the resulting documents are stored in MongoDB to-

gether with basic metadata and processing status. Implementation details are described in Chapter 6.

3. **Protest classification.** Each scraped article is labelled at the article level as `PROTEST` or `NOT_PROTEST` using a transformer based zero-shot classifier, producing both a label and a confidence score. The reader can find more details in section 5.2.

4. **Downstream analysis.** The protest subset is used for the main analyses in this thesis: sentiment (VADER), exploratory topic modelling, and temporal comparisons across outlets and time periods.

The number of items retained at each stage is summarised in Table 8.1. Throughout the results chapter, descriptive coverage analyses use the predicted `PROTEST` set ($N = 3{,}046$), while sentiment analyses use the subset with valid sentiment scores ($N = 2{,}823$).

Pipeline diagram and data flow

Figure 5.1 summarises the data flow from URL retrieval to analysis outputs.

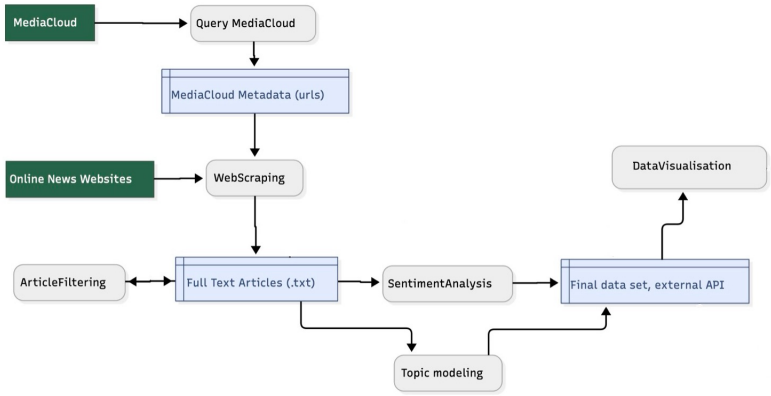

Figure 5.1: Pipeline diagram and data flow.

5.2 Binary classification: protest vs. non protest

Task definition and unit of analysis

The classification task is defined at the level of individual articles. For each scraped article the code decides whether its main topic is a concrete protest event. The label set is:

- 1 = **protest**: the article's main focus is a collective, public action (demonstration, march, rally, strike, riot, blockade, occupation, picket, etc.) in which people express political or social claims. The article should describe the event itself (who, where, why, what happened) or its immediate unfolding (e.g. clashes, arrests, dispersal).
- 0 = **not protest**: the article main focus are not protests

This coding rule is used both for manual labelling and to interpret the output of the classifier. It is deliberately narrower than "any contentious or chaotic event", which means that some crowd disorder like illegal raves or riots without clear claims are not counted as protest in the PEA sense.

Pre filtering and sampling

To reduce the number of obviously irrelevant articles before classification, Some filtering is applied. Articles whose URLs clearly indicate sports, lifestyle, entertainment, restaurant reviews or similar soft news are removed at the URL stage and never scraped.

From this pool, a manually labelled dataset was created, stored in a MongoDB collection called `sample_texts`, and exported to CSV for modelling. The evaluation set used for the zero-shot classifier contains around 85 articles, of which 23 are protest articles and the remainder are non protest. Articles were sampled randomly, so the evaluation set is imbalanced, reflecting the fact that protest coverage is a minority in the total corpus.

Manual labelling protocol

Manual labelling follows the binary codebook above. For each sampled article I read the full text and assigned a label:

- 1 if the article's main focus is a protest event as defined above;
- 0 otherwise.

Borderline cases, for example articles that mix event description with a lot of historical or contextual discussion, were labelled according to whether the protest event is central to the story.

For unclear cases, like articles that describe an event but also include lots of background, they were labelled based on whether the protest event is the main focus of the story.

Evaluation and threshold selection

The zero-shot classifier outputs, for each article, a confidence score p that the article belongs to the "protest" class. Instead of using the default 0.5 threshold, the code searched for a threshold that performs well on the manually labelled evaluation set. This is implemented in the script `threshold.py`.

Let p be the model's probability for the protest label. For a given threshold t was assigned a predicted label:

$$\hat{y} = \begin{cases} 1 & \text{if } p \geq t \\ 0 & \text{if } p < t. \end{cases}$$

For each threshold t between 0 and 1 the code computes the confusion matrix and derive standard classification metrics. In addition to accuracy, precision, recall and F_1, F_β with $\beta = 0.5$ was calculated:

$$F_\beta = (1 + \beta^2) \cdot \frac{\text{precision} \cdot \text{recall}}{\beta^2 \cdot \text{precision} + \text{recall}}.$$

With $\beta < 1$, $F_{0.5}$ places more weight on precision than recall. In this context, that choice is deliberate: for downstream analysis it is better to have a relatively clean set of protest articles, which translates in fewer false positives, even if some genuine protest articles are missed. Therefore the threshold t that maximises $F_{0.5}$ on the evaluation set, was selected.

Applying the classifier to the full corpus

Using the selected threshold, the code applies the zero-shot classifier to all scraped articles that pass the initial URL and section filters. The predictions are written back into the main MongoDB collection (`Texts`), which stores for each article:

- the predicted protest label (0/1)
- the associated confidence score

Having the confidence score gives the possibility of changing the threshold if needed without having to go through the whole classification again.

5.3 Sentiment analysis

Sentiment analysis is used to characterise the tone of protest coverage over time and across outlets. This project uses VADER (Valence Aware Dictionary and Sentiment Reasoner), a lexicon and rule based sentiment tool designed for English.

For each article classified as protest related, VADER is applied to the full text to obtain:

- a compound sentiment score in the interval $[-1, 1]$, where more negative values indicate more negative language and more positive values indicate more positive language;
- component scores for negative, neutral, and positive proportions of the text.

At this stage, sentiment is treated as the overall tone of the article's language. In other words, it captures whether the text sounds broadly positive or negative as a whole, rather than showing attitudes towards specific actors. This is an important limitation, because news articles often contain mixed language—for example, they may describe protesters sympathetically while describing disruption or violence negatively. A single sentiment score cannot separate these different targets, so it should be seen as a rough summary of tone. Accordingly, the sentiment results are interpreted as the overall tone of coverage, not as sentiment towards protesters or other actors involved. In later chapters,these article level sentiment scores are combined to see how average sentiment changes by newspaper and over time.

5.4 Topic modelling and thematic analysis

Beyond the question of whether an article is about protest, it is important to understand what kinds of protest it covers, for example; cost of living protests, Black Lives Matter, anti lockdown demonstrations, Palestine solidarity marches, or climate activism.

To explore this, the study applies an unsupervised topic modelling approach to the subset of articles classified as protest related. The aim is to identify groups of articles that discuss similar protest issues and frames.

Unsupervised topic modelling with BERTopic

In the implemented pipeline, topic modelling is performed using BERTopic. First, the input is restricted to articles labelled as `PROTEST`, and basic text cleaning is performed, including whitespace normalisation and removal of repeated patterns. Very short texts are excluded to reduce noise.

BERTopic represents each article using sentence embeddings, like Sentence-Transformer `all-MiniLM-L6-v2`, and then clusters these embeddings to form topics. To make topics interpretable,`CountVectorizer` was used with English stopwords and both unigrams and bigrams `ngram_range=(1,2)`, `min_df=5`. The main tuning choices are the minimum topic size,`min_topic_size=40`, and whether to reduce the number of topics `nr_topics=None` in the current implementation.

For each article, BERTopic outputs a topic label and if doable a topic confidence score based on the model's topic probability distribution. Results are summarised by computing topic shares over time, binned quarterly, and by newspaper outlet. Because topic models require interpretation, the resulting topics are analysed qualitatively by inspecting the most representative terms and example documents for each topic.

5.5 Temporal analysis

A core aim of this thesis is to compare how UK newspapers covered protest before, during, and after the COVID-19 pandemic. To do this, the dissertation analyses how protest related reporting changes over time.

First, the study measures how much protest coverage there is by counting the

number of articles classified as protest over time. The analysis computes these counts at different time resolutions. Weekly counts are useful for identifying peak weeks in protest coverage. Monthly or yearly counts provide a clearer long run picture. Weekly peak detection is automated and exported to CSV for further inspection.

Second, the analysis compares broad pandemic periods. Following the project's definitions, the timeline is split into:

- **Pre-COVID:** 2020-01-01 to 2020-03-11
- **COVID:** 2020-03-11 to 2022-02-24
- **Post-COVID:** 2022-02-24 to 2024-12-31

This study uses 2020-03-11 since it was the pandemic declaration date and 2022-02-24 as a practical boundary because UK COVID restrictions were largely lifted around this period, making it a reasonable operational cut point for 'post-COVID' media coverage. The dissertation uses these periods to summarise how the volume of protest coverage differs across phases and across outlets.

Third, the study examines how tone changes over time. Using the VADER compound score already computed per article, there is a comparison on the sentiment across the three COVID periods and by outlet. In addition to comparing averages, the dissertation also compares the full sentiment distributions across periods.

Overall, the temporal analysis is descriptive: it focuses on trends, comparisons between outlets, and contrasts between COVID phases.

5.6 Evaluation strategy

Each major component of the pipeline raises different evaluation questions:

- **Binary classification.** For the protest vs. non protest classifier, the main metrics are precision, recall, and F-scores for the protest class, along with overall accuracy and comparison to a simple majority class baseline. Threshold selection uses $F_{0.5}$ to favour precision, as described in Section 5.2. Error analysis is performed by manually reviewing false positives and false negatives and identifying systematic patterns.

- **Sentiment.** For sentiment scores, there is no simple ground truth. Instead, evaluation focuses on face validity: checking whether obviously positive or negative articles receive appropriate scores, and testing whether average sentiment behaves sensibly across outlets and over time.
- **Topics.** For topics, evaluation is likewise qualitative and analytical. Useful checks include whether discovered themes align with peaks in protest coverage correspond to well known events.
- **Temporal patterns.** For time series evaluations, we are going to check if peaks in protest coverage correspond to protest waves in the period.

A more detailed discussion of model training, testing and evaluation, including qualitative examples, is given in Chapter 7.

6 Implementation

While the previous chapter focuses on the conceptual design of the pipeline, this chapter describes how it is implemented in practice. Section 6.1 summarises the development environment and key libraries. Section 6.2 explains how scraping and cleaning scripts are organised.

6.1 Development environment

The project is implemented in Python 3.9.6. Development and testing were carried out on a personal machine using a virtual environment to manage dependencies. The main libraries used include:

- **Data handling and utilities:** `pandas` for tabular data, `numpy` for basic numerical operations, and `tqdm` for progress bars.
- **Web scraping:** `requests` for HTTP requests and `newspaper3k` usind `fun_scrap3.py` for article text extraction.
- **Database access:** `pymongo` for interacting with MongoDB, along with `python-dotenv` to load database credentials and configuration from environment variables.
- **NLP models:** Hugging Face `transformers` for the zero-shot classifier and additional utilities from the Hugging Face ecosystem for tokenisation and training.
- **Sentiment:** VADER sentiment analysis, accessed either via `nltk` or the standalone `vaderSentiment` package.
- **Topic:** BERTopic model, from `berttopic`, also used SentenceTransformer from `sentence_transformers` and CountVectorizer from `sklearn.feature_extraction.text`.

- **Plotting and analysis:** `matplotlib` and optionally `seaborn` for figures in the analysis stage.

6.2 Data pipeline implementation

MediaCloud export and first query

The pipeline begins with a MediaCloud CSV export of candidate articles. The script `FirstQuery.py` standardises the export format, applies the inclusion and section filtering rules defined in Chapter 4, and produces a cleaned list of URLs for scraping.

The output is saved as `URLS_clean.csv`, which serves as the input to the scraping stage.

Scraping articles

The main scraping logic lives in `fun_scrap3.py`, which defines a function that:

1. takes a URL as input;
2. fetches the HTML (with basic error handling for HTTP errors such as 403 and 404);
3. passes the HTML to `newspaper3k` for parsing and text extraction;
4. returns the article text.

To scrape articles at scale and store them in the database, `scrape_to_mongo.py` reads the cleaned URL export (`URLS_clean.csv`), removes duplicates, and skips URLs that have already been successfully processed in MongoDB. For each URL, the script:

- adds a short random delay to reduce rate limiting and avoid "robotic" request patterns;
- calls the scraping routine `scrape_and_text(url, title)` from `fun_scrap3.py` to extract the article text;

- checks the output and flags failures; `SCRAPE_FAILED` or `EMPTY_TEXT`, recording an error status when needed;
- upserts the result into the MongoDB collection `Texts`, using the URL as `_id`, storing metadata such as `paper`, `title`, and `publish_date`, along with other fields like `status`, `time_enqueued`, and `time_scraped`.

To avoid duplicate scraping, the URL is used as identifier in MongoDB, the `_id` field. If an article already exists in the database, the script can skip rescraping or only update fields that were previously missing.

6.3 Model training and classification scripts

Zero-shot classifier implementation

The main classifier is implemented in `hf_class.py`, which wraps a Hugging Face zero-shot classification pipeline. Given an article text, it calls the model with two candidate labels such as:

- PROTEST_LABEL = "a concrete real world protest event"
- OTHER_LABEL = "something else"

and returns the model's probabilities for each label. The script standardises this into:

- a binary label (1 for protest, 0 for non protest) using a configurable threshold;
- a numerical protest score $p \in [0, 1]$;
- raw model outputs for debugging.

`run_hf.py` applies this classifier to the full scraped corpus stored in the `Texts` collection. It iterates over articles that do not yet have a prediction, sends the text to the classifier, and writes the results back into the same document. Progress is tracked with `tqdm` and simple logging, and the script can be restarted if it stops part way through.

Threshold sweeps and evaluation

Threshold optimisation and evaluation are handled in `threshold.py`. This script:

- reads the labelled evaluation sample from `sample_texts`;
- retrieves the corresponding zero-shot scores p from MongoDB or from a CSV export;
- loops over possible thresholds t between 0 and 1;
- for each t, computes the confusion matrix and metrics; accuracy, precision, recall, F_1, and $F_{0.5}$;
- identifies the threshold that maximises $F_{0.5}$.

The chosen threshold is then hard coded or stored in a configuration variable so that `run_hf.py` uses it consistently when classifying the full corpus.

7 Model Training, Testing and Evaluation

7.1 Problem formulation

In this dissertation, protest detection is framed as a binary text classification task at the level of individual news articles. Each observation is a UK news article, represented by its headline and full text. The goal is to estimate whether an article main focus is a concrete protest event.

Formally, for each article i we observe a pair of texts (h_i, t_i), where h_i is the headline and t_i is the article body. The classifier outputs a probability

$$f : (h_i, t_i) \to p_i \in [0, 1],$$

where p_i is the model's estimated probability that the article contains a concrete protest event.

A decision threshold $\tau \in (0, 1)$ is applied to obtain a predicted label

$$\hat{y}_i = \begin{cases} 1, & \text{if } p_i \geq \tau, \\ 0, & \text{if } p_i < \tau. \end{cases}$$

The threshold is selected to prioritise high precision, to have fewer false positives, because false positives would inflate estimated protest coverage in later analysis.

Once evaluated, the classifier is applied to the full corpus to:

- estimate the volume of protest coverage over time for each outlet;
- compare protest coverage before, during, and after the COVID–19 period;
- analyse differences in protest coverage across ideologically diverse newspapers.

7.2 Data for training and evaluation

7.2.1 Corpus for inference

The full dataset was collected using a MediaCloud search and a web scraping pipeline, and stored in MongoDB. Articles were retrieved using a combination of time filtering and protest related keyword filtering.

The final scraped corpus contains $N = 12{,}496$ articles from three outlets:

- Daily Mail: 8,680 articles
- The Guardian: 2,790 articles
- Evening Standard: 1,026 articles

This distribution is imbalanced and reflects the availability of articles.

7.2.2 Automatic labelling output

After applying the final classification threshold (Section 7.5.2), 12,496 articles received a model prediction: 3,046 were classified as protest and 9,256 as non protest. The remaining 194 articles were not classified due to scraping errors or missing orinsufficient text.

7.2.3 Human labelled evaluation sample

Because the final approach uses a zero–shot model, performance was evaluated using a small human labelled sample.

A random sample of 85 articles was selected from the database and manually checked to create a human label indicating whether the article reports a concrete protest event.

In the $N = 85$ evaluation set, the human labels contain:

- 23 protest articles
- 62 nonprotest articles

7.3 Models

7.3.1 Baselines

A simple baseline for comparison is the majority class baseline, which predicts all articles as non protest. On the evaluation sample, this baseline achieves an accuracy of:

$$\frac{62}{85} \approx 0.729,$$

but it achieves $F_1 = 0$ for the protest class because it never predicts a protest.

7.3.2 Transformer based model (zero-shot, Hugging Face)

The final classifier is a zero–shot transformer model implemented using the Hugging Face `zero-shot-classification` pipeline. The underlying pretrained model is `facebook/bart-large-mnli`.

Each article is classified using two candidate labels:

- ``a concrete real world protest event''
- ``something else (no specific protest event)''

The model returns a score for each label. This study uses the score for the protest label as $p_i = P(\text{PROTEST})$, and applies a threshold τ to determine the final binary prediction.

7.4 Training setup

No gradient based training or fine tuning was needed for the zero-shot classifier. The model was used "as is" for inference, and the main adjustable parameters are the decision threshold τ, and the descriptions of the labels.

7.5 Evaluation protocol

7.5.1 Metrics

Performance was evaluated using standard classification metrics computed on the human labelled evaluation sample, like Accuracy, Precision, Recall, F_1, $F_{0.5}$. The protest class is the substantively important class. Precision is especially important in this dissertation because false positives would inflate estimates of protest coverage in later chapters.

7.5.2 Decision threshold selection

The zero-shot model outputs a probability like score for the protest label. Instead of using the default threshold $\tau = 0.5$, multiple thresholds were tested to manage the trade off between precision and recall.

Thresholds were swept from 0.05 to 0.90 in steps of 0.01. Using the confidence score, for each threshold, the code recalculated the confusion matrix and metrics. The final threshold was selected by maximising $F_{0.5}$, which weights precision more heavily than recall.

The optimal threshold in this sweep was:

$$\tau = 0.65.$$

7.6 Results on the evaluation set

7.6.1 Confusion matrix

Using $\tau = 0.65$, the confusion matrix on the $N = 85$ evaluation sample is:

	Predicted Protest	**Predicted Non Protest**
Actual Protest	11 (TP)	12 (FN)
Actual Non Protest	10 (FP)	52 (TN)

Table 7.1: Confusion matrix on the human labelled evaluation sample ($N = 85$) at threshold $\tau = 0.65$.

The model correctly identified 11 out of 23 protest articles, while missing 12 protest articles. It incorrectly labelled 10 non protest articles as protest.

7.6.2 Performance metrics

At $\tau = 0.65$, performance on the protest class is:

- Accuracy: 0.741
- Precision: 0.524
- Recall: 0.478
- F_1: 0.500
- $F_{0.5}$: 0.514

The baseline is a majority class model, which always predicts the most common lable in the same, in this case `NOT_PROTEST`. In this the baseline achieves Accuracy ≈ 0.729, but $F_1 = 0$. The zero-shot model therefore improves accuracy modestly and, more importantly, achieves non zero performance on the protest class.

7.7 Error analysis

A brief qualitative review of model errors suggests several recurring patterns.

Some false positives occur when articles describe large crowds, unrest, or disorder that is not a protest event. Examples include:

- reports about riots or disorder in prisons, which contain similar vocabulary to protest coverage;
- crowd events such as the Qatar World Cup opening ceremony, where dense crowds in a confined space can resemble protest descriptions;
- disaster coverage, where language about anger, chaos, looting, or people gathering can be misinterpreted as a protest signal.

False negatives tend to occur when protests are described in a calm or indirect way, or when protest is not emphasised in the headline. For example, a May Day article about Pro Palestine demonstrations in London was missed. More generally, peaceful demonstrations can be harder for the model to detect when the article focuses on arrests, traffic disruption, or political context rather than the protest event itself.

These patterns suggest that the model relies heavily on lexical cues and may confuse protest with other forms of collective gatherings or disorder.

7.8 Limitations and implications for downstream analysis

This classification approach has several important limitations.

First, performance was evaluated on a relatively small human labelled sample. This limits confidence in precise performance estimates and makes it difficult to test performance separately by outlet or time period.

Second, the threshold selection and performance reporting were conducted on the same evaluation sample due to limited annotations. This may slightly overestimate performance compared to a fully independent held out test set.

Third, the corpus is imbalanced across outlets, The Daily Mail dominates the dataset.

Finally, a small number of articles could not be classified due to scraping errors or missing/short text. These articles are excluded from the protest coverage estimates.

Despite these limitations, the model is applied consistently across all outlets and time periods. For this dissertation, the predictions are therefore used as approximate indicators of protest coverage, suitable for relative comparisons across time and outlets, rather than as perfect ground truth.

In the following chapter, the model predictions for the full corpus are used to analyse trends in protest reporting and differences across UK newspapers.

8 Findings and Discussion

8.1 Corpus overview

This chapter reports findings from an automated pipeline designed to analyse how UK newspapers report on protest. A protest event may take place inside or outside the UK, as long as it is reported by the selected UK outlets.

Table 8.1: Pipeline totals and drop off across processing stages.

Stage	Count
Documents stored in MongoDB	12,506
Successfully scraped	12,496
Successfully classified	12,302
Predicted `PROTEST`	3,046
Predicted `NOT_PROTEST`	9,256
`PROTEST` with valid sentiment	2,823

Throughout this chapter, results are reported on the relevant subset of the corpus. Unless otherwise stated, protest coverage analyses use the `PROTEST` set ($N = 3{,}046$), while sentiment analyses use the subset with valid sentiment scores ($N = 2{,}823$; Table 8.1).

8.1.1 Data pipeline and attrition

The MediaCloud query returned 59,742 URLs across many outlets. After filtering down to three outlets and removing URLs that were not relevant news articles, the dataset was reduced to 12,506 URLs. From these, 12,496 articles were successfully scraped and stored in MongoDB. A total of 12,302 articles were successfully classified, while 194 were failed or skipped at the classification stage.

The protest/not-protest classifier, applying the final threshold of 0.65, produced 3,046 items labelled as `PROTEST` and 9,256 labelled as `NOT_PROTEST`.

8.1.2 Outlet composition

The final protest corpus is imbalanced across outlets. The Daily Mail has 1,953 protest articles, The Guardian has 547, and The Evening Standard has 546. Of the 546 Evening Standard articles, 223 were scraping error pages, so they are excluded from sentiment analysis, leaving 323 usable articles. This imbalance affects how strongly outlets can be compared.

The outlets were selected for two reasons: they were feasible to scrape reliably at scale, and they span a commonly discussed ideological spectrum in the UK media, from The Guardian (centre left) to The Daily Mail (right wing), with The Evening Standard included as a London focused title [Smith, 2017].

8.1.3 Protest articles by year

Table 8.2 shows protest article counts by year and outlet. The largest volume occurs in 2021 with 957 articles, followed by 2020 with 784. Counts fall in 2022 and 2023, with a modest increase in 2024.

Table 8.2: Counts of `PROTEST` articles by year and outlet.

Year	Daily Mail	Evening Standard	The Guardian	Total
2020	531	125	128	784
2021	695	117	145	957
2022	281	74	95	450
2023	205	90	99	394
2024	241	140	80	461
Total	1,953	546	547	3,046

8.2 Protest coverage over time

8.2.1 Pre, during, and post COVID comparison

To compare coverage around the pandemic timeline, three time periods were defined. The boundary for the start of COVID is the WHO pandemic declaration on 11 March 2020. The end of the COVID period is set as 24 February 2022, when the UK government removed key remaining legal restrictions such as the legal requirement to selfisolate after a positive test. These boundaries are used as

a practical way to group coverage into interpretable phases rather than as a claim that COVID related effects ended on a single day.

- **Pre-COVID:** 2020-01-01 to 2020-03-11.
- **COVID period:** 2020-03-11 to 2022-02-24.
- **Post-COVID:** 2022-02-24 to 2024-12-31.

Because the pre-COVID window is short, counts and sentiment estimates for this period are based on a small n and should be interpreted cautiously.

Table 8.3 shows that protest coverage is much higher during and after COVID than in the pre-COVID window, which is short and therefore produces a small n.

Table 8.3: Counts of `PROTEST` articles by COVID period (overall and by outlet).

Outlet	Pre-COVID	COVID	Post-COVID	Total
All outlets	88	1,741	1,217	3,046
Daily Mail	67	1,228	664	1,953
Evening Standard	10	241	295	546
The Guardian	17	272	258	547

8.2.2 Peaks and notable periods

The time series shows that protest coverage is not spread evenly across time. Instead, a small number of high attention moments produce clear spikes in the number of articles. The script `weekly_protests_peaks.py`, counts how many articles were labelled `PROTEST` in each week. The 10 weeks with the highest counts are shown in Table C.4.

Several of these peak weeks are consecutive, so it makes sense to group them into short periods of sustained high coverage. Table 8.4 summarizes these periods.

Table 8.4: Periods with higher counts of `PROTEST` articles.

Period	Protest articles
2021-01-05 to 2021-02-01	275
2020-05-26 to 2020-06-15	225
2024-07-29 to 2024-08-12	102

To understand what was driving these peaks, `check_art_title.py` was used to inspect article titles during each period. The first period, 2021-01-05 to 2021-02-01, is mainly linked to the US Capitol attack and the protests around it. In this dissertation, this is treated as protest related coverage because it involved a large public mobilisation with a clear political aim, even though it also included violence. Example titles include **"Trump supporters storm US Capitol building in clashes with police"** and **"Officers who took selfies inside Capitol during riots are fired"**.

The second period, 2020-05-26 to 2020-06-15, starts right after the killing of George Floyd. This triggered a major wave of Black Lives Matter protests, mainly in the USA, but also in the UK and many other countries. This period also overlaps with the early and most uncertain phase of COVID-19, when restrictions were strict and widely debated, so some protest coverage is also linked to lockdown rules. Example titles include **"Police in riot gear spray tear gas and fire rubber bullets at protest over George Floyd's death"** and **"Woman blinded by police rubber bullet at BLM march has 'no regrets' "**.

Finally, the third period, 2024-07-29 to 2024-08-12, is linked to antiimmigration unrest in the UK following a stabbing in Southport in which three children were killed. In media reports, far right groups spread claims about the suspect (including claims that he was a Muslim asylum seeker), which contributed to wider disorder and racist attacks [Bello, 2025]. Example titles include **"Violent clashes between protesters and police in Southport after stabbings"** and **"REVEALED: The Russian linked fake news website which fuelled lies about the Southport stabbings and sparked violent protests"**.

8.3 Sentiment and framing

Sentiment was measured using a lexicon based approach (VADER). This provides a simple signal of positive/negative language, but it does not directly measure political bias or normative support for protesters. What we can do is compare different sentiment over the different newspapers and their differences during before and after COVID; in the Table C.5 we can see these differences:

All of this data has been collected from the file `sentiment_mean_by_outlet_and_covid_period.csv`, this file was created using the code from `vader_covid.py`.

VADER sentiment scores for protest related articles in this dataset are strongly negative, with medians close to -1 across newspapers and COVID periods. This suggests that protest reporting is usually written in negative language, often fo-

cusing on conflict, disruption, policing, or violence. However, many scores are extremely close to -1, which means the results are "compressed" at the negative end of the scale see figure B.2. This can happen with VADER because it is a word list method and long news articles repeat negative words, pushing the score towards the minimum. For this reason, in this thesis VADER is best used as a relative measure: it is useful for comparing sentiment between newspapers and across time periods, but it should not be treated as an exact measure of the emotional tone of each individual article. We can see the distribution in this the figure B.2.

Across outlets, sentiment distributions differ substantially. The Daily Mail and The Guardian show consistently extreme negative medians, approximately -0.99, across periods, indicating that at least half of protest related articles in these outlets are scored as highly negative by VADER. By contrast, The Evening Standard exhibits markedly less negative sentiment during and after COVID, means around -0.43 ; medians around -0.66, suggesting a less uniformly negative framing. During the COVID period, The Guardian's mean sentiment is more negative than The Daily Mail's, but the two outlets converge post COVID. Pre COVID comparisons for The Evening Standard and Guardian should be interpreted cautiously due to small sample sizes.

8.3.1 Sentiment labels by outlet

Across all outlets, the majority of protest articles are classified as negative in sentiment. The proportions are:

- **Daily Mail (1,953):** 1,691 negative, 256 positive, 6 neutral.
- **The Guardian (547):** 488 negative, 57 positive, 2 neutral.
- **Evening Standard (323):** 245 negative, 68 positive, 10 neutral.

A key interpretation point is that "negative" language can reflect many things violence, conflict, tragedy, or critical political context, not only hostility toward protesters.

8.3.2 Mean sentiment by COVID period

Mean VADER compound scores are strongly negative overall, with only modest differences over time:

- **Pre-COVID:** mean -0.744 ($n = 85$)
- **COVID:** mean -0.724 ($n = 1{,}634$)
- **Post-COVID:** mean -0.636 ($n = 1{,}104$)

By outlet, The Evening Standard is less negative on average than the other two outlets during COVID and post-COVID. However, the pre-COVID Evening Standard estimate is based on only 7 articles and should not be over interpreted.

Overall, the sentiment results suggest that tone is consistently negative across outlets and periods. Differences exist, but they are not large, and outlet comparisons are limited by the strong corpus imbalance in favour of The Daily Mail.

8.4 Issues and topics

Topic modelling was implemented using BERTopic in `topic_modelling.py`. This produced topic assignments for individual articles and topic share summaries over time and across outlets.

Since BERTopic is not perfectly deterministic, the clustering and dimensionality reduction steps can be sensitive to random initialisation and to the order in which documents are retrieved from the database, so repeated runs may produce small differences in the topics. In this project, the model identified between 11 and 13 topics (see `topic_info.csv`), and Figure B.3 shows how the highest volume topics change over time.

The strongest peaks in the topic timeline align with the high attention periods identified earlier: Black Lives Matter, the US Capitol related episode, and UK anti immigration unrest. Smaller clusters reflect other international protest waves covered by UK media, including protests linked to the Gaza war and the war in Ukraine.

One additional cluster reflects scraping artefacts for example, pages where the scraper captured an error or access message instead of article text. This is substantively useful because it highlights that topic modelling can also function as a quality control step by identifying systematic scraping failures.

9 Conclusions

This dissertation set out to test whether an automated NLP pipeline can support PEA on UK newspaper coverage, and whether the resulting outputs are useful for comparing protest reporting before, during, and after COVID-19. The project treated the news article as the primary unit of analysis and focused on building an end to end workflow from URL discovery to scraping, protest relevance classification, and later descriptive analyses.

9.1 Summary of contributions and main findings

The project makes four main contributions.

(1) A practical, reproducible pipeline for corpus construction. The dissertation implements a workflow that (i) queries MediaCloud for protest related candidates, (ii) filters to selected outlets and removes out of scope sections, (iii) scrapes article text at scale, and (iv) stores raw and derived fields in a documented MongoDB schema. This addresses a recurring constraint in PEA research: turning large volumes of heterogeneous news content into a queryable research data set [Hutter, 2014].

(2) An operational definition of "protest article" and a deployable classifier. This project sets a clear rule for deciding if an article is about protest: the article counts as `PROTEST` when its main focus is a real, specific protest event. The study then applies this rule using a transformer based zero shot classifier, which can label texts without training a new model on a large hand labelled dataset. This makes it possible to filter a very large corpus in a consistent way. However, accuracy is only moderate: the model works well as a first step to find likely protest articles, but it still makes mistakes and cannot be treated as ground truth. This matches earlier warnings in the literature that automated event data needs clear validation and transparent reporting [Lorenzini and Kriesi, 2021, Abdurahman, 2025].

(3) A structured basis for comparative reporting analysis across outlets and time. Once classification is applied consistently, the outputs can be aggregated into interpretable descriptive results: protest coverage volume over time, comparisons across COVID periods, and cross outlet contrasts. The time series analysis shows that "protest coverage" is highly peaked: a small number of high attention windows dominate the volume, and these peaks can be driven by international events that UK outlets choose to foreground.

(4) A sober assessment of what is and is not feasible within an MSc scale project. The pipeline succeeds at retrieval, storage, and protest relevance filtering, but struggles with more ambitious goals like reliable event attribute extraction without a dedicated event level annotation effort. This reinforces a central point in PEA: newspapers are not structured "event logs", and their narrative form creates real limits for fully automated extraction [Hutter, 2014, Radiukiewicz and Dubrow, 2024].

9.2 Final answers to the research questions

How effectively can an automated NLP pipeline distinguish genuine protest reports from other news articles in a large corpus?

The pipeline can distinguish protest focused articles from non protest articles to a moderate degree. It is strong enough to reduce a large candidate corpus to a smaller set that is plausibly protest relevant, enabling downstream analysis at scale. However, the errors are still consistent. False positives often include crowd trouble, disasters, or unrest stories that use protest language, while false negatives happen when protest is mentioned indirectly or described in long and calm official language. In short, the classifier is useful for filtering and prioritisation, not for fully automated PEA without human validation [Lorenzini and Kriesi, 2021, Abdurahman, 2025].

How does the sentiment and framing of protest coverage vary across ideologically diverse newspapers and over time?

Using lexicon based sentiment as a proxy for tone, protest coverage is consistently skewed negative across outlets and across the pandemic timeline. This does not imply uniform hostility to protesters: negative language may reflect violence, disruption, tragedy, policing, or conflict oriented news values rather than normative judgement. Differences between outlets are present but are smaller

than might be expected from ideological accounts alone. On top of that, interpretation is constrained by outlet imbalance in the corpus and the mismatch between "sentiment" and "framing" as theorised in the social movements literature [Benford and Snow, 2000, Susánszky, 2022].

To what extent do topic patterns in protest reporting shift across the pandemic timeline?

The topic model suggests that protest reporting is shaped by a small number of high attention issues rather than many stable themes. Across outlets, the same big topics appear in all newspapers—especially the U.S. Capitol riot, Black Lives Matter, and UK anti immigration protests. While smaller clusters cover protests against COVID-19 restrictions mostly in the UK and EU and protests linked to international conflicts such as Gaza and Ukraine. Across the pandemic timeline, the main changes seem to follow major events rather than a single strong "COVID topic": restriction related protests are more visible during the pandemic years, but later coverage shifts more toward post pandemic crises and war related mobilisation, with some low frequency international topics appearing more weakly and being harder to interpret. Overall, differences across outlets look more like differences in how much they cover each issue than completely different topic agendas.

9.3 Limitations and threats to validity

Several limitations shape how the findings should be interpreted.

Selection and coverage bias. The corpus is not a census of all protest reporting. It depends on MediaCloud's indexing and on a keyword based retrieval strategy, which will miss protests described without the chosen vocabulary and may include articles that mention protest only incidentally. The analysis also covers only three outlets, chosen for accessibility and ideological contrast; this limits generalisability to the UK media system as a whole.

Scraping and measurement noise. Web scraping yields imperfect text: boilerplate, truncated pages, and layout changes introduce noise. This can affect classification and downstream sentiment/topic analysis. While basic length filtering mitigates the worst cases, it does not eliminate subtle extraction artefacts.

Classifier evaluation limits. The labelled evaluation sample is small and class imbalanced. In addition, threshold selection and performance reporting were

conducted on the same sample due to limited annotations, so performance may be optimistic.

Sentiment as a proxy for framing. VADER sentiment provides a simple and reproducible tone measure, but it is not a direct operationalisation of "media framing". It cannot tell who the negative language is aimed at, such as protesters, authorities, or the events around them. It also reacts strongly to common news language about conflict and tragedy, not to the writer's position.

Article-level rather than event level analysis. Because event attribute extraction was not implemented reliably, the dissertation does not produce a clean event dataset comparable to ACLED style records [ACLED, 2015]. The conclusions therefore speak to media coverage of protest, not to the true underlying distribution of protest events.

9.4 Recommendations and future work

The most valuable next steps are those that increase validity without exploding project complexity.

(1) Improve protest detection with more labelled data and an active learning loop. A modest expansion of manual labels would enable in domain fine tuning and a more defensible evaluation design. Active learning can target borderline items where the model is uncertain, improving sample efficiency.

(2) Move from document level to sentence/segment level modelling. Many errors arise because articles mix protest description with context, commentary, and unrelated narrative. Segmenting articles and classifying segments could reduce both false positives and false negatives and would also support later extraction.

(3) Pilot event attribute extraction as a constrained task. A feasible intermediate step is to extract only a small set of high salience attributes on a curated subset of high confidence protest articles, with manual validation. This would test what is realistically extractable from newspaper narratives without over claiming full automation [Radiukiewicz and Dubrow, 2024, Abdurahman, 2025].

9.5 Concluding remarks

Overall, the dissertation demonstrates that an end to end automated workflow for protest focused media analysis is feasible and useful as scalable filter and descriptive monitoring tool. It also shows, in a concrete way, why fully automated PEA remains difficult: news articles are selective, narrative, and ideologically framed sources rather than structured event records [Hutter, 2014]. The most defensible role for NLP/LLM methods in this setting is therefore hybrid: automation for retrieval, filtering, and descriptive summarisation, paired with targeted human validation for claims about protest dynamics and media bias [Lorenzini and Kriesi, 2021, Abdurahman, 2025].

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A Tools and software used in the project

This appendix summarises the main tools used to build the dataset and run the analysis, and explains why each tool was chosen. The goal is to make the workflow easier to understand.

Code repository

The full source code for this project is available on GitHub: https://github.com /Birkbeck/msc-project-source-code-files-24-25-eledelaf

Use of AI

I am an international student and English is not my first language. I wrote the full dissertation and used generative AI as a a tool to improve wording, fix grammar, and make sentences clearer. I did not copy AI output directly, instead, I used it as feedback and then rewrote the text myself. AI was not used to create or alter the academic content like methods, analysis, results, or conclusions.

A.1 Programming environment

The full pipeline was implemented in Python.

A.2 MediaCloud

MIT MediaCloud was used as the main source to collect candidate news articles. MediaCloud was chosen because it provides a large archive of online news,

supports keyword based queries, and allows exporting results as structured files. This makes it suitable for creating an initial candidate set at scale before downloading full article text.

A.3 Data filtering and cleaning

The initial MediaCloud export was processed with a custom Python script to:

- restrict to the chosen date range and outlets
- remove invalid or missing URLs
- exclude URLs that clearly point to irrelevant sections

This step was included to reduce noise and avoid spending time scraping content that is outside the scope of the study. Standard Python data tools were used here because they are simple, transparent, and easy to rerun.

A.4 Web scraping

Full article text was collected using `newspaper3k`, `from newspaper import Article, Config`. This library was chosen because it provides a practical, end to end way to extract the main article content from news URLs: it downloads the page, parses the HTML, and applies built in heuristics to identify the headline and main text while ignoring most navigation, adverts, and others.

In this project, `Config` was used to control scraping behaviour and `Article` was used to perform the download, parsing, and text extraction. Compared with writing custom HTML rules for each newspaper, this approach is faster to implement and easier to scale to thousands of URLs.

A limitation is that automatic extraction is not always reliable for modern news sites, especially when pages are paywalled, heavily dynamic, or blocked by anti bot protections. For this reason, the scraper records a clear status for each URL and stores error information when extraction fails. This is important because scraping failures are not random and may vary across outlets and time periods, which can introduce bias into later analysis.

A.5 MongoDB

Scraped articles and derived outputs were stored in MongoDB. MongoDB was chosen for two main reasons:

- **Flexible structure:** the project stores both raw fields like URL, title, text, publish date and fields created later like classification outputs.
- **Pipeline tracking:** each document can store processing status flags, making it easier to rerun steps safely and to audit where and why data was lost.

In addition, MongoDB was initially selected because I expected the datasets to be very large and did not want to store them locally on my computer. In practice, the final datasets were smaller than expected, so the project could also have been implemented with local files. Because the data is relatively small, it can be downloaded and stored locally using the files provided in the `8.MongoDB` folder. The database was accessed from Python using the `pymongo` library.

A.6 Protest / Not-Protest classification

To identify articles whose main focus is a concrete protest event, the project used a transformer based zero-shot classifier from the Hugging Face ecosystem (model: `facebook/bart-large-mnli`). This approach was chosen because it avoids the need for large labelled training data and can be applied consistently across outlets and time periods. The classifier outputs a confidence score, which was converted into a binary decision using a threshold.

Zero-shot classification is a method where a pre trained language model assigns labels to a text without being trained on labelled examples from the specific project. Instead of learning from a protest labelled training dataset, the model uses its general language understanding and the meaning of the label descriptions in this case "a concrete real world protest event", to estimate how well each label matches the article. This makes it useful as a first pass filter when labelled data is limited, but it also has limitations: results depend on how the labels are worded and accuracy can vary across topics, outlets, and writing styles.

A.7 Threshold selection and human validation

A small human coded sample was used to support threshold selection and quality checks. Using that sample, and the confidence score, we select the threshold that maximises the metric. This step is included to make the classification rule explicit and to provide transparency about expected error. Even with threshold tuning, some misclassification remains, so downstream findings are interpreted with appropriate caution.

A.8 Sentiment analysis

For protest articles, sentiment was estimated using Valence Aware Dictionary and Sentiment Reasoner. VADER is a rule based, lexicon based sentiment method designed for English text. It works by using a predefined dictionary of words and expressions, where each entry is associated with a typical sentiment polarity and strength (positive, negative, or neutral). When VADER processes a text, it looks up sentiment bearing words in this lexicon and combines them into overall scores.

The output includes proportions of positive, negative, and neutral content, and a single compound score that summarises the overall sentiment of the text.

VADER was chosen because it is fast, easy to apply at scale, and widely used for English sentiment scoring, which makes it suitable for this data set. However, because it is lexicon based, it has known limitations for news and political language: it can be strongly influenced by specific negative terms, it does not reliably capture sarcasm or complex context, and it does not measure sentiment toward specific actors.

A.9 Topic modelling

To summarise recurring themes in protest reporting, the project used `BERTopic`. BERTopic was selected because it combines:

- transformer embeddings
- clustering
- keyword based topic representations

This makes it suitable for large text corpus where topics may shift over time. Topic modelling is exploratory by nature: topics depend on model settings and the available text, and topic labels are ultimately a human interpretation of the top words and representative documents.

A.10 Data analysis and plotting

Core analysis and visualisation were done using standard Python libraries. `Pandas` and `NumPy` where used for counts, time series, group comparisons and `matplotlib` for figures.

These libraries were chosen because they are reliable, widely used, and support fully reproducible analysis pipelines.

B Figures

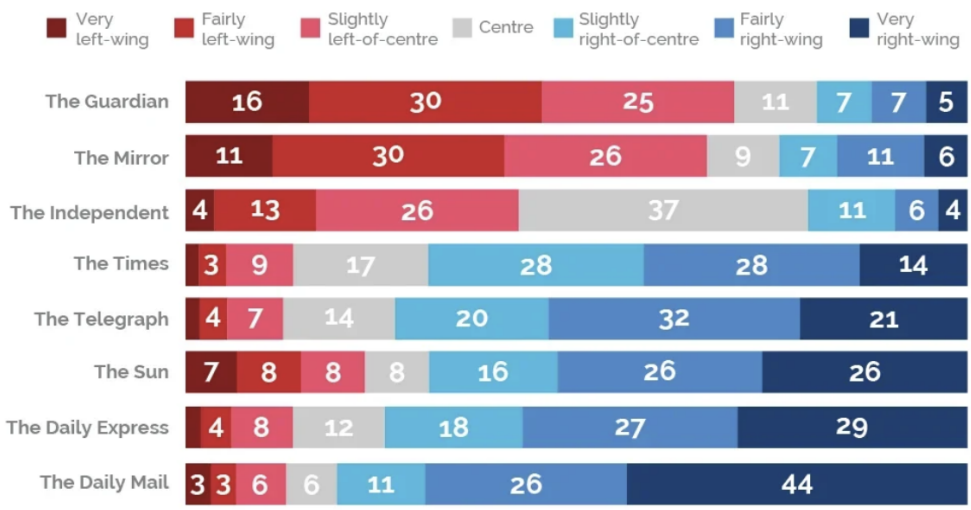

Figure B.1: Perceived ideological position of selected UK newspapers (YouGov).

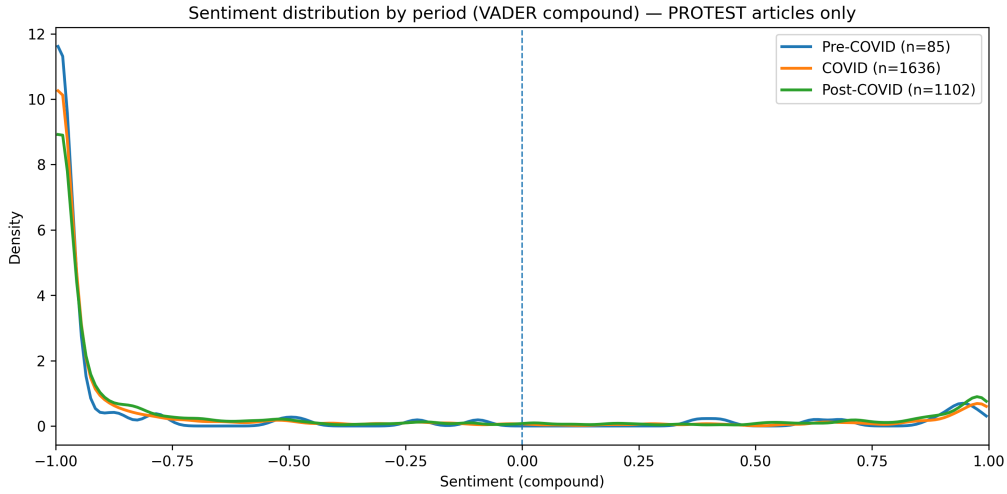

Figure B.2: Sentiment distribution

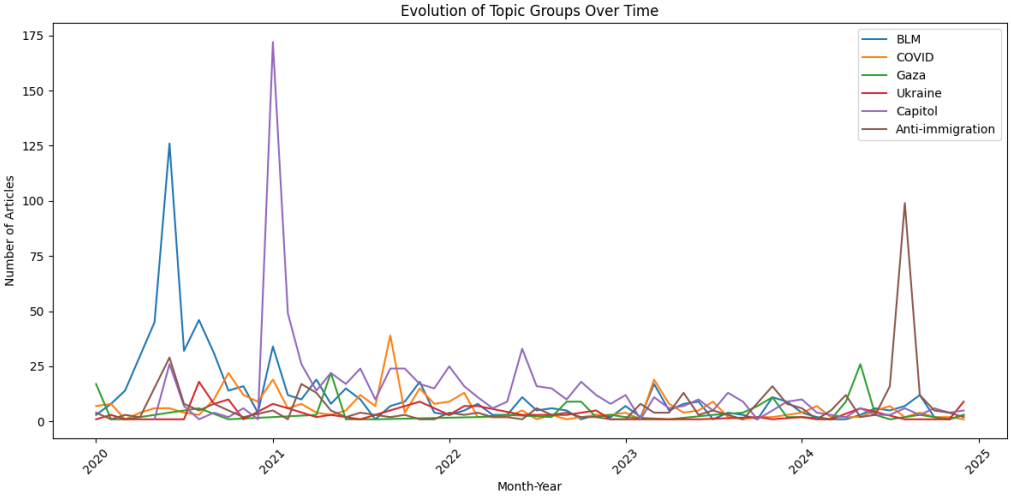

Figure B.3: Number of protest articles about a certain topic vs time publish.

C Tables

Table C.1: Schema of the `Texts` collection in MongoDB

Field	Type	Description
`_id`	string	Unique identifier for the document, set to the article URL.
`publish_date`	date	Date when the article was published (from MediaCloud).
`status`	string	Scraping status for the article, e.g. ``done'', ``pending'', ``failed'', ``error''.
`time_enqueued`	datetime	Timestamp when the URL was queued for scraping.
`title`	string	Article headline, taken from MediaCloud or the scraped page.
`text`	string	Main body text of the article as extracted by `newspaper3k`.
`time_scraped`	datetime	Timestamp when scraping completed successfully.
`paper`	string	Newspaper label derived from the URL.
`hf_confidence`	float	Confidence score returned by the Hugging Face classifier for the predicted label.
`hf_label`	integer	Predicted binary label from the classifier: `0` for "not protest", `1` for "protest".
`hf_label_name`	string	Human readable label name corresponding to `hf_label`.
`hf_model`	string	Identifier or name of the Hugging Face model used for classification.
`hf_reason`	string	Additional diagnostic information from the classification stage, used internally for debugging.
`hf_status`	string	Status of the classification step, e.g. ``ok'', ``error'', ``failed_or_skipped''.

Table C.2: Schema of the `sentiment` of the texts label as "PROTEST" in MongoDB

Field	Type	Description
`label`	string	Status of the sentiment analysis ``positive'', ``negative'', ``neutral''
`neg`	float	A value between 0 and 1 that represents how much negative sentiment there is in the text.
`neu`	float	A value between 0 and 1 that represents how much neutral sentiment there is in the text.
`pos`	float	A value between 0 and 1 that represents how much positive sentiment there is in the text.
`compound`	float	Combined score based on `neg`, `neu`, and `pos` representing overall sentiment.

Table C.3: Schema of the `sample_texts` collection in MongoDB

Field	Type	Description
`_id`	string	Article URL, used as unique identifier.
`title`	string	Article headline.
`text`	string	Scraped article text.
`human_label`	integer	Manual binary label: `0` for "not protest", `1` for "protest".

Table C.4: Top 10 weeks by number of `PROTEST` articles.

Week	Protest articles
2020-06-01 to 2020-06-08	108
2021-01-11 to 2021-01-18	102
2021-01-05 to 2021-01-11	82
2020-05-26 to 2020-06-01	73
2021-01-18 to 2021-01-25	54
2024-08-05 to 2024-08-12	52
2024-07-29 to 2024-08-05	50
2020-06-08 to 2020-06-15	44
2021-01-25 to 2021-02-01	37
2020-06-22 to 2020-06-29	35

Table C.5: sentiment mean by newspaper and covid period

Paper	N	COVID period	mean	median	std
Daily Mail	61	Pre-COVID	-0.86	-0.99	0.45
Daily Mail	1228	COVID	-0.73	-0.99	0.60
Daily Mail	664	Post-COVID	-0.68	-0.99	0.67
Evening Standard	7	Pre-COVID	0.30	0.62	0.73
Evening Standard	134	COVID	-0.43	-0.62	0.55
Evening Standard	182	Post-COVID	-0.43	-0.66	0.58
The Guardian	17	Pre-COVID	-0.77	-0.99	0.51
The Guardian	272	COVID	-0.82	-0.99	0.47
The Guardian	258	Post-COVID	-0.68	-0.98	0.64